Stroke

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Thromboinflammation in Acute Stroke: From Pathophysiological Challenges to Emerging Therapeutic Opportunities

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ABSTRACT: Despite major advances in intravenous thrombolysis and endovascular thrombectomy, nearly half of patients with acute ischemic stroke fail to achieve functional recovery even after technically successful recanalization. The recanalization-reperfusion gap—the discordance between angiographic vessel opening and tissue-level perfusion recovery—has brought thromboinflammation, the pathological interplay of coagulation and innate immunity, to the forefront of stroke biology. Neutrophil extracellular traps, platelet-leukocyte aggregates, complement activation, and vessel wall inflammation render clots resistant to lysis, promote microvascular obstruction and the no-reflow phenomenon, and amplify ischemia-reperfusion injury. These insights reframe stroke not solely as a problem of reopening arteries, but as an inflammatory disorder in which thrombus biology, microcirculatory flow, and inflammatory injury determine outcome. In this review, we outline the key thromboinflammatory challenges—recanalization resistance, no-reflow, and reperfusion injury—and discuss therapeutic opportunities that emerge from this framework. These include enzymatic neutrophil extracellular traps–targeting, adjunctive anti-inflammatory agents, biomarker-guided precision approaches, and novel device technologies that exploit or mitigate thrombus biology. Collectively, these strategies support a paradigm shift in stroke care: from procedure-focused reperfusion to biologically informed interventions that integrate vascular and immune determinants of outcome.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: extracellular traps ■ microcirculation ■ reperfusion injury ■ thrombectomy ■ thromboinflammation

Acute ischemic stroke most often results from the occlusion of a cervical or cerebral artery, leading to focal cerebral hypoperfusion and a time-dependent cascade of metabolic failure, excitotoxicity, oxidative stress, and ultimately cell death—culminating in the formation of an infarct.[1–3] Timely reperfusion halts infarct progression and preserves salvageable tissue.[4,5] As such, recanalization of the occluded vessel, whether pharmacologically through intravenous thrombolysis or mechanically via endovascular thrombectomy, is the cornerstone of acute ischemic stroke treatment.

Despite major advances in recanalization strategies over the past decade,[6] a substantial proportion of patients continue to experience poor outcomes. Nearly half of those who achieve successful recanalization fail to regain functional independence.[7] This discrepancy highlights that vessel opening does not guarantee tissue salvage, raising the need to examine biological mechanisms beyond angiographic success.

To understand these poor outcomes despite vessel recanalization, it is essential to distinguish between 3 related but distinct concepts. Recanalization refers to the angiographic reopening of an occluded vessel—a macrovascular phenomenon assessed by imaging modalities such as the modified Thrombolysis in Cerebral Infarction scale. Reperfusion, in contrast, describes the actual restoration of blood flow at the tissue and microvascular level, which may or may not occur even after successful

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recanalization. The no-reflow phenomenon represents the most extreme manifestation of this discordance: despite angiographically confirmed vessel patency, microvascular perfusion fails due to capillary obstruction by leukocyte-platelet aggregates, distal microemboli, endothelial dysfunction, and vasogenic edema compressing the microcirculation. Recognizing this recanalization-reperfusion gap is fundamental to understanding why angiographic success does not guarantee clinical recovery and why targeting downstream thromboinflammatory mechanisms is critical.

One key driver of this recanalization-reperfusion gap is thromboinflammation, which refers to the interplay between coagulation and innate immunity (Figure 1).[7–10] At both the primary thrombus site and the downstream microvasculature, neutrophil extracellular traps (NETs), platelet-leukocyte aggregates, complement activation, and endothelial dysfunction promote a highly thrombogenic and inflammatory environment.[11] In particular, NETs contribute to the no-reflow phenomenon,[12–14] directly promoting thrombus propagation, thrombus stabilization, and recanalization resistance.[15–18] Recognizing the pervasive role of thromboinflammation reframes stroke care—shifting the focus from purely procedural macrovascular recanalization to a biologically informed paradigm that addresses both large-vessel patency and microvascular health.

In this review, we explore the challenges and therapeutic opportunities emerging from this evolving understanding—emphasizing how thromboinflammatory mechanisms offer novel avenues for targeted intervention.

THROMBOINFLAMMATION CHALLENGES

Recanalization Resistance

Despite being the primary target of acute stroke therapy, clot removal remains incomplete or unsuccessful in a significant proportion of patients in centers with optimized mechanical thrombectomy protocols. Intravenous thrombolysis achieves early recanalization in only 20% to 30% of large-vessel occlusions, whereas endovascular thrombectomy fails to achieve recanalization in ≈10% to 20% of cases, even in high-volume centers with experienced interventionalists.[6,19]

A major explanation lies in the evolving architecture and biology of thrombi over time.[16] Histopathologic studies classify thrombi as red blood cell–rich, fibrin-rich, or mixed-type, each with distinct mechanical and biochemical characteristics.[20] Red blood cell–rich thrombi are softer and more amenable to aspiration, whereas fibrin-rich clots adhere more tightly to the endothelium, often requiring stent retrievers or combined approaches.[21,22] Regardless of initial composition, thrombi progressively become more resistant through the recruitment of platelets, leukocytes, and NETs, reinforcing their structure and reducing their susceptibility to lysis.[11]

NETs, in particular, form a DNA-histone scaffold that creates an outer shell, stabilizing the thrombus and hindering thrombolysis.[11] Over time, thrombi may become densely packed and cross-linked, forming a complex matrix that mechanical devices can only partially engage, leading to fragmented or incomplete retrieval.[23] Another

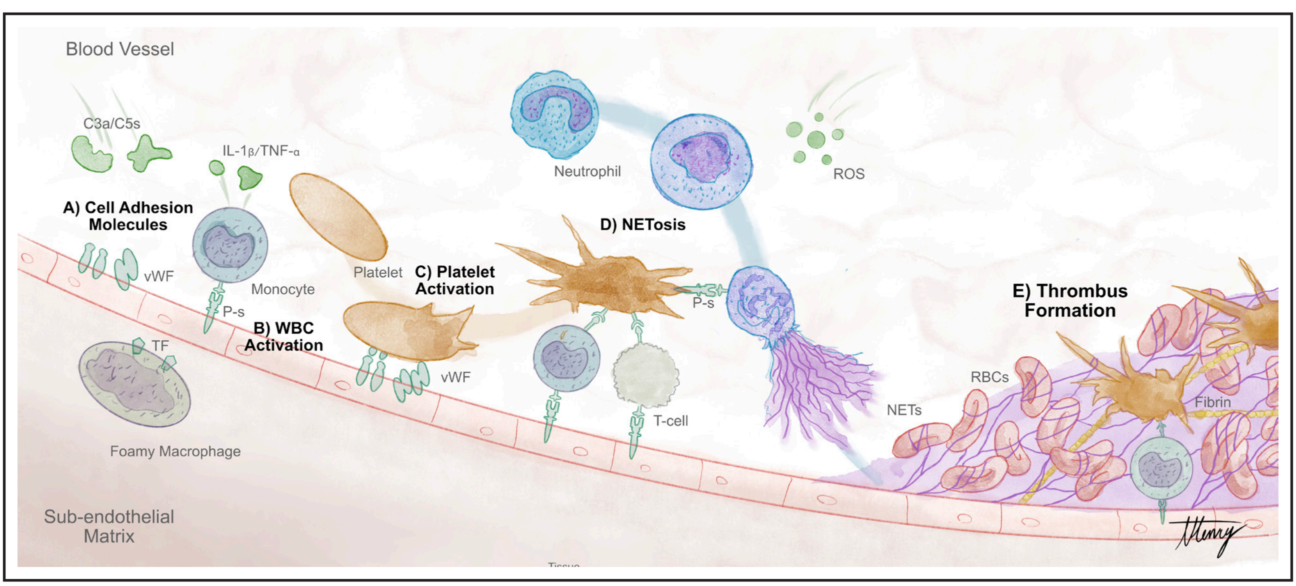

Figure 1. Thromboinflammatory cascade.
A, Complement proteins (C3a, C5a) and subendothelial foamy macrophages induce endothelial adhesion molecules (VWF [von Willebrand Factor], P-selectin [P-s]). **B**, White blood cell (WBC) activation: monocytes release proinflammatory cytokines (IL [interleukin]-1β/TNF-α [tumor necrosis factor-α]), and macrophages express TF (tissue factor), initiating coagulation. **C**, Adhesion molecules drive platelet activation, platelet-leukocyte aggregation, neutrophil recruitment, and reactive oxygen species (ROS) generation, culminating in (**D**) NETosis: neutrophils release DNA-histone extracellular traps (neutrophil extracellular traps [NETs]). **E**, NETs, platelet-leukocyte aggregates, and fibrin entrap red blood cells (RBCs), consolidating thrombus formation.

proposed mechanism is clot compaction caused by the blood-hammer effect of pulsatile flow proximal to the thrombus, which increases wall tension and further resists aspiration or stent retriever forces.[24,25] Compaction also decreases internal voids within the clot, limiting the surface area for thrombolytic agents. The dynamic nature of pulsatile flow additionally alters the delivery and clearance of clotting factors and circulating cells, promoting further compaction.

NETs have therefore emerged as central effectors of clot resistance. Released by receptor-activated or dying neutrophils (NETosis), they act as a physical scaffold that stabilizes thrombi and traps circulating platelets and red blood cells.[26] Clinical studies confirm that NET-rich thrombi are less responsive to thrombolysis and more difficult to retrieve mechanically.[16] Alongside NETs, platelet-leukocyte aggregates,[27] proinflammatory cytokines,[28] and complement components[29] further stabilize clots and enhance vessel wall adhesion, creating a highly prothrombotic environment. In this setting, thrombi are not just passive vascular obstructions but active immunothrombotic entities.

A better understanding of thrombus composition and age could help tailor therapy—such as choosing adjuvant treatment to enhance thrombolysis, or choosing aspiration versus stentriever-first approaches.[30] For instance, the Clotild guidewire combines impedance-based microsensors and machine learning to characterize clot composition.[31] Such real-time thrombus characterization could inform not only mechanical strategy selection but also guide adjunctive therapies targeting specific thromboinflammatory components, such as NET-degrading enzymes for NET-rich clots or anti-inflammatory agents for complement-activated thrombi, thereby optimizing recanalization success and downstream microvascular reperfusion. However, these strategies remain investigational and warrant prospective validation in biomarker-guided trials.

No-Reflow Phenomenon

Even when angiographic recanalization is technically successful, tissue reperfusion is not guaranteed. This mismatch is increasingly attributed to the no-reflow phenomenon—initially described in myocardial infarction[12] and now well-documented in the cerebral circulation (Figure 2).[32–34] No-reflow refers to the failure of microvascular reperfusion despite the reopening of upstream occluded arteries. In ischemic stroke, this phenomenon results from a combination of microvascular obstruction, endothelial dysfunction, and inflammatory damage. Several mechanisms compromise the microcirculation: (1) capillary plugging by activated leukocytes and platelet aggregates, where staling neutrophils interact with platelets to form primary occlusive scaffolds,[35–37] (2) thrombotic debris from distal embolization of the primary thrombus,[33] and (3) external compression of capillaries caused by vasogenic edema.[38]

Therefore, angiographic indices of complete recanalization, such as a Thrombolysis in Cerebral Infarction score of 2c or 3, may overestimate true tissue-level reperfusion. Advanced imaging modalities, including computed tomography and magnetic resonance imaging perfusion acquired after the endovascular procedure, can visualize arteriolar and capillary-level flow and reveal such mismatches.[39] Studies show that a perfusion–angiography mismatch predicts poor functional outcome even in technically successful endovascular thrombectomy cases.[40–42] Yet, these tools are not routinely integrated into intraprocedural workflows or reperfusion grading.

Emerging evidence suggests that adjunctive intra-arterial thrombolysis after mechanical thrombectomy may address residual microvascular obstruction in no-reflow scenarios.[43] The CHOICE trial (Intraarterial Alteplase Versus Placebo After Mechanical Thrombectomy) demonstrated that intra-arterial alteplase administration after thrombectomy significantly reduced residual perfusion deficits assessed by quantitative angiographic analysis, even in patients who achieved complete angiographic recanalization (modified Thrombolysis in Cerebral Infarction scale score 2c-3).[44] These findings suggest that pharmacological targeting of residual microemboli and thromboinflammatory products in the distal microvasculature may complement mechanical macrovascular recanalization, potentially narrowing the recanalization-reperfusion gap in selected patients with persistent tissue-level hypoperfusion.

Ischemia-Reperfusion Injury: Oxidative Stress After Successful Recanalization

In contrast to the no-reflow phenomenon, where microvascular perfusion fails despite vessel patency, when tissue-level reperfusion does occur, it can paradoxically trigger oxidative injury—a phenomenon known as ischemia-reperfusion injury.[45] A key driver is mitochondrial succinate accumulation during ischemia. Under normal aerobic conditions, succinate is oxidized by complex II (succinate dehydrogenase) in the electron transport chain. However, during ischemia, oxygen deprivation halts oxidative phosphorylation while anaerobic metabolism continues, causing succinate to accumulate to levels 5- to 10-fold higher than baseline. This accumulation results from the reversal of succinate dehydrogenase activity and the breakdown of purine nucleotides through the malate-aspartate shuttle. On reperfusion and reoxygenation, the accumulated succinate is rapidly oxidized by complex II, driving an intense electron flux that overwhelms complex I. This creates a reversal of electron flow from complex II back to complex I (reverse electron transport), where electrons react with molecular oxygen to generate a massive burst of superoxide and other reactive oxygen species (ROS) within the first minutes of

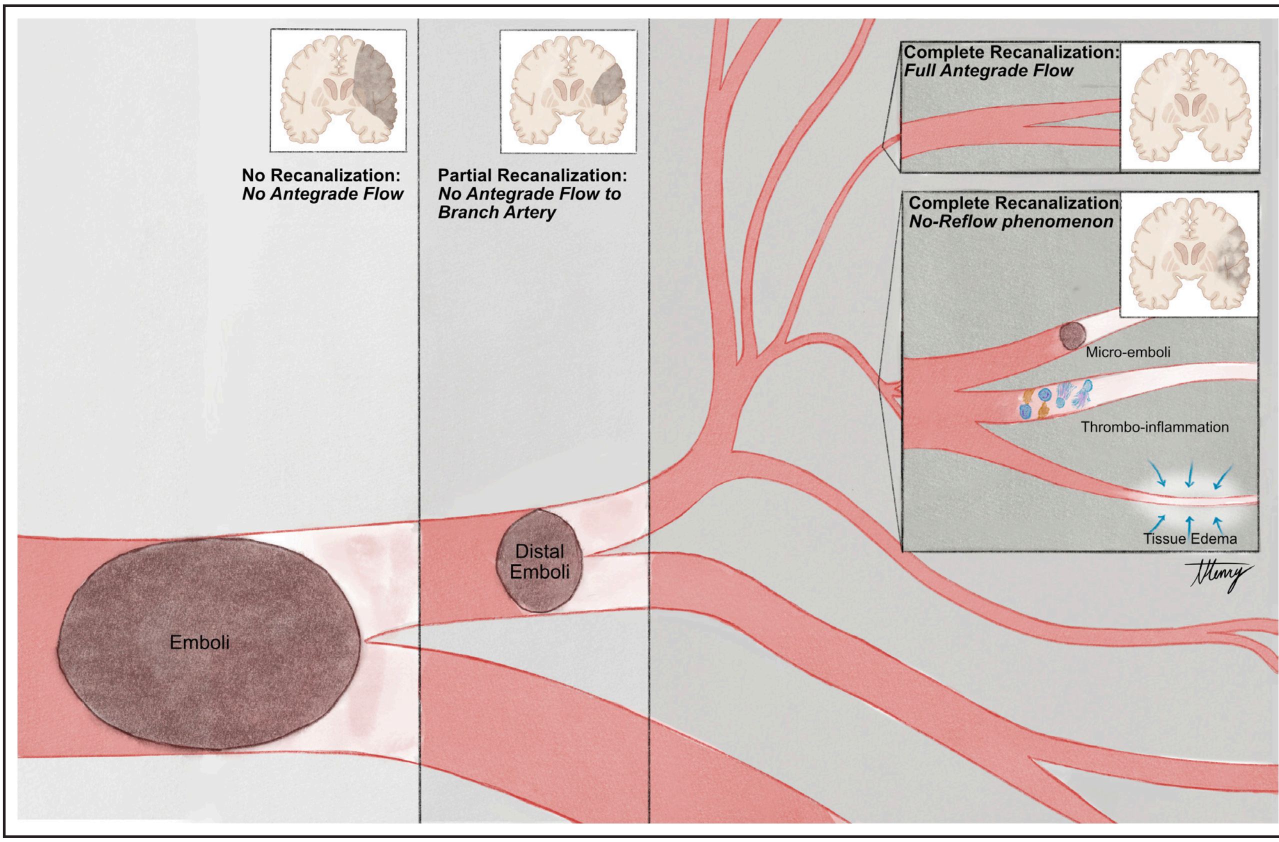

Figure 2. The no-reflow phenomenon.

From **left** to **right**: (1) persistent large-vessel occlusion with no anterograde flow; (2) partial recanalization with distal thrombus fragmentation into a medium-distal vessel and no anterograde flow to branch artery; (3) complete recanalization but persistent distal no-flow caused by microemboli, thromboinflammation, or tissue vasogenic edema impairing arteriolar and capillary perfusion (no-reflow phenomenon); and (4) complete recanalization with full anterograde flow.

reperfusion. This ROS burst initiates a cascade of reperfusion injury.[46]

ROS directly disrupt cellular membranes and DNA, and trigger the opening of mitochondrial permeability transition pores, leading to mitochondrial dysfunction, calcium overload, and eventual energy failure. In parallel, ROS activate inflammasome pathways, particularly NLRP3 (NLR family pyrin domain–containing 3), amplifying neuroinflammation and cytokine release.[47] This injury is aggravated by NET components—especially citrullinated histones— that worsen endothelial dysfunction, activate the complement cascade, and promote blood-brain barrier breakdown.[48] Preexisting or stress-induced hyperglycemia further exacerbates these inflammatory processes by accelerating lactic acidosis and amplifying oxidative stress.[49] ROS also impair microvascular function through vasoconstriction, blood-brain barrier disruption, and vasogenic edema, exacerbating the no-reflow phenomenon.[50]

Critically, oxidative stress and inflammatory damage are deeply interconnected processes that amplify one another. ROS-induced mitochondrial dysfunction activates damage-associated molecular patterns, which in turn prime NLRP3 inflammasome assembly and trigger inflammatory cytokine release. Conversely, inflammatory mediators—particularly TNF-α (tumor necrosis factor-α) and IL (interleukin)-1β—stimulate NADPH oxidase activity in microglia and infiltrating neutrophils, generating additional ROS bursts that perpetuate oxidative injury. This positive feedback loop between oxidative and inflammatory pathways creates a self-sustaining cycle of tissue damage that extends far beyond the initial ischemic insult.

Although no-reflow and ischemia-reperfusion injury represent distinct pathophysiological states, they share common thromboinflammatory mediators that create a mechanistic continuum. In partial or heterogeneous reperfusion—where some microvascular territories achieve perfusion while others remain obstructed—oxidative and inflammatory cascades initiated in reperfused regions (ROS, cytokines, NETs) propagate into adjacent no-reflow zones, exacerbating microvascular dysfunction and expanding the zone of injury. Moreover, the same thromboinflammatory processes (neutrophil activation, endothelial dysfunction, and complement activation) that cause no-reflow also prime tissues for greater susceptibility to reperfusion injury if flow is eventually restored. Notably, injury is particularly severe in patients with prolonged ischemia or systemic inflammation, where these mechanisms are amplified. Severe hypoperfusion, especially when prolonged or accompanied by systemic factors such as hyperglycemia, substantially increases the risk of complications such as hemorrhagic transformation and cerebral edema.[49,51,52]

The concept of an ischemic core region so severely damaged that reperfusion becomes not only nonbeneficial but potentially harmful by precipitating these ischemic-reperfusion complications is now described as the leaky core.[49] Current stroke treatments—focused on achieving large-vessel recanalization—do not address these downstream tissue damage mechanisms that follow the initial ischemic event and persist even after vessel patency is restored, making neuroprotection against oxidative and inflammatory injury an urgent unmet need.

Blood and Imaging Biomarkers of Thromboinflammation

Although a wide range of inflammatory and thromboinflammatory biomarkers have shown strong associations with stroke severity and outcome, few are currently implemented in clinical workflows. On the blood side, circulating markers, such as matrix osteopontin,[53] IL-6, and ADAMTS13 (a disintegrin and metalloproteinase with a thrombospondin type 1 motif, member 13),[54] cell-free DNA,[55] and NET-associated components like citrullinated histone H3[56] are elevated in acute ischemic stroke and correlate with infarct size, hemorrhagic transformation, and poor clinical outcome.[57] Similarly, complement activation products, such as C3a and C5a, reflect innate immune activation and predict blood-brain barrier disruption and edema.[58]

On the imaging side, advanced techniques, such as flat-panel perfusion, arterial spin labeling, and high-resolution vessel wall magnetic resonance imaging, can detect tissue at risk, vessel wall inflammation, microvascular dysfunction, and plaque instability.[59,60] Despite these promising data, the lack of assay standardization, limited validation in large cohorts, and logistical challenges still prevent their clinical integration.

Thromboinflammation in Atherothrombotic and Device-Induced Cerebral Ischemia

Although most thromboinflammatory research focuses on embolic stroke, atherothrombotic events—including spontaneous plaque rupture in intracranial atherosclerotic disease[61] or carotid artery disease,[62] as well as stent-induced arterial injury[63]—present distinct and complex inflammatory challenges. In these settings, disruption of the fibrous cap exposes subendothelial matrix and lipids, triggering a wound-healing–like hemostatic response. When neutrophils and proinflammatory macrophages fail to clear lipid debris and necrotic core material, the process remains unresolved, resulting in persistent inflammation and thrombus propagation—akin to chronic plaque environments in coronary atherosclerosis.[64]

The clinical use of intracranial stents further compounds the problem. Each stent breaches the endothelium, causing multiple microdissections, whereas the persistent foreign surface sustains endothelial activation and leukocyte adhesion. The microdissections trigger acute inflammatory responses, whereas the foreign body itself sustains chronic inflammation. Together, these processes promote sustained leukocyte recruitment and activation, culminating in exaggerated fibrotic tissue response that drives neointimal hyperplasia and in-stent restenosis.[65] Moreover, compared with self-expanding stents, balloon-expandable stents pose a heightened risk of dissection or perforation during implantation due to the lack of adventitia and a thin media.[66] Finally, although drug-eluting stents (eg, rapamycin-eluting) have been used off-label, concerns about neurovascular toxicity persist despite reassuring animal studies.[67] The combination of mechanical endothelial disruption, chronic inflammation, and foreign body response underscores the need for device designs tailored to neurovascular thromboinflammatory biology.

Interindividual Variability in Thromboinflammation

A growing body of evidence suggests that biological sex, hormonal milieu, genetic background, and environmental factors modulate key pathways of thromboinflammatory activation, helping explain interindividual differences in stroke severity, thrombus composition, and treatment response—and supporting the case for personalized inflammatory profiling in precision stroke medicine.[68,69]

Although women over 60 years of age bear a higher lifetime risk of stroke compared with men,[70] partly attributable to loss of estrogen-mediated vascular protection postmenopause, men experience higher stroke incidence at younger ages—a pattern that underscores the importance of understanding sex-specific thromboinflammatory mechanisms across the lifespan.[71] Estrogen inhibits NET formation via antioxidant mechanisms and modulates endothelial activation through NO and prostacyclin signaling. In contrast, androgens can promote pro-NETotic phenotypes by increasing ROS production and PAD4 (peptidyl arginine deiminase 4) activation. Clinically, postmenopausal women exhibit heightened systemic inflammation and increased NET biomarkers.[72] Moreover, genetic polymorphisms in coagulation and immune response pathways—including complement receptors, Fcγ receptors, and inflammasome components—shape the intensity and profile of thromboinflammatory responses, driving variability in NETosis, complement activation, and vessel wall inflammation.[73–75]

THERAPEUTIC OPPORTUNITIES

Targeting NETs to Improve Recanalization and Reperfusion

NETs are increasingly recognized as critical mediators of thromboinflammation, contributing to thrombus resistance to recanalization and to the downstream no-reflow phenomenon (Figure 3). Their dense DNA-histone scaffold interlinks fibrin, platelets, and leukocytes, enhancing clot rigidity, reducing r-tPA (recombinant tissue-type plasminogen activator) penetration, and promoting adherence to the vessel wall via interactions with VWF (von Willebrand Factor) and fibrin.[76–78]

Two approaches have been investigated: (1) pharmacological NETs degradation as an adjunctive therapy to enhance recanalization response to thrombolysis and (2) leveraging NETs to improve mechanical thrombectomy. In preclinical stroke models, enzymatic degradation of NETs using DNase I (deoxyribonuclease I) enhanced thrombolysis, improved recanalization, reduced infarct volume, and improved neurological outcomes. In models of large-vessel occlusion, the combination of r-tPA and DNase I accelerated clot lysis and improved vessel patency.[76] These findings have prompted 2 randomized clinical trials, which are currently evaluating the use of recombinant human DNase I (dornase alfa) in acute ischemic stroke due to large-vessel occlusion: (1) an open-label phase II trial assessing intravenous dornase alfa in patients eligible for intravenous thrombolysis and thrombectomy (NETs-target: URL: https://www.clinicaltrials.gov; Unique identifier: NCT04785066) and (2) a Bayesian, dose-finding umbrella trial (EXTEND-IA DNase [Improving Early Reperfusion With Adjuvant Dornase Alfa in Large Vessel Ischemic Stroke; URL: https://www.clinicaltrials.gov; Unique identifier: NCT05203224]) aimed at identifying an optimal dornase alfa dose based on achieving substantial reperfusion or absence of retrievable thrombus at the first angiogram.

Beyond degradation, another approach leverages the structural resilience of NETs to enhance mechanical

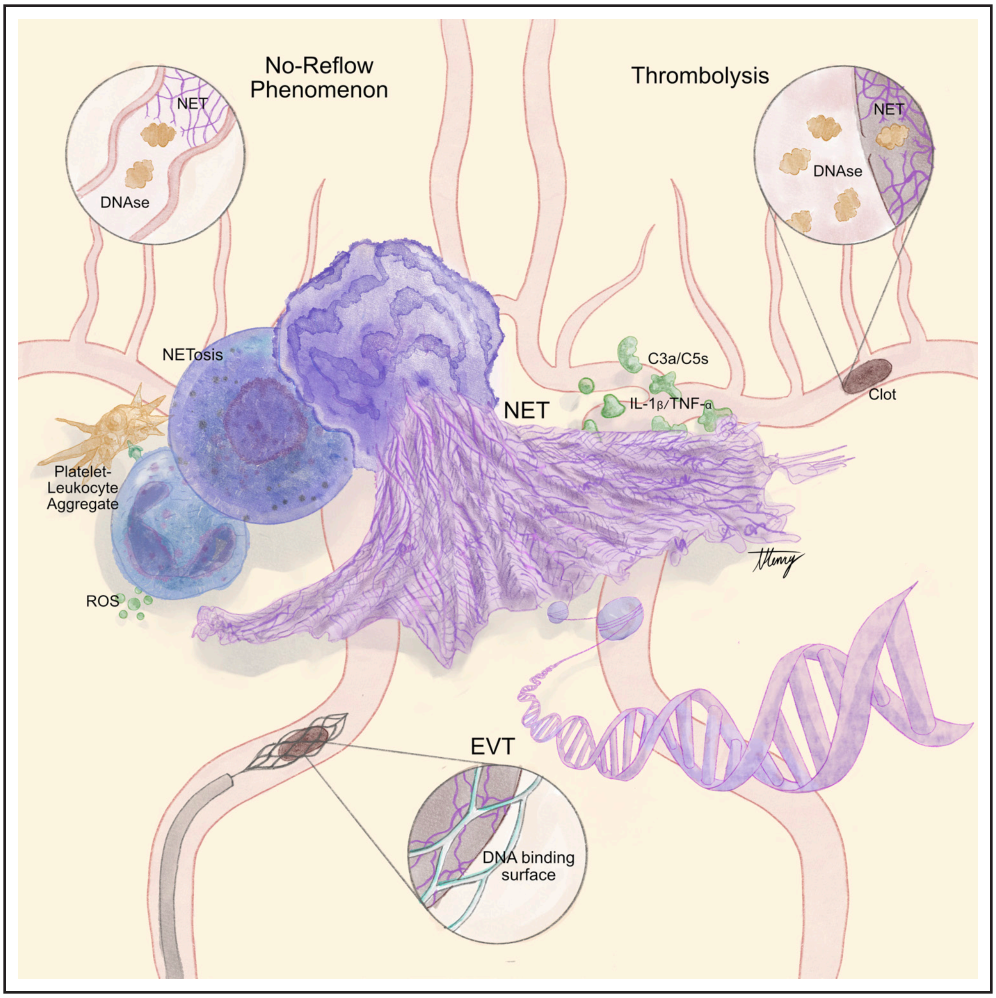

Figure 3. Therapeutic strategies targeting neutrophil extracellular traps (NETs).
Therapeutic strategies include: (**A**) reducing no-reflow; (**B**) enhancing thrombolysis (eg, DNase I adjuncts); and (**C**) device surface approaches with DNA-binding ligands to improve of thrombi and their fragments. EVT indicates endovascular thrombectomy; IL, interleukin; ROS, reactive oxygen species; and TNF-α, tumor necrosis factor-α.

thrombectomy. Given that NETs act as a fibrous scaffold interlinking fibrin, platelets, and red blood cells, they also contribute to thrombus cohesiveness and surface adherence. These features not only complicate retrieval[79] but also provide anchoring points for device engagement. A notable example is the work by Skarbek et al,[80] who anchored the DNA-binding antimalarial piperaquine onto the surface of clinical-grade stent retrievers via click chemistry. This surface functionalization increased clot engagement, achieving ≈50% more first-pass captures than bare-metal devices and reduced the number of embolic fragments by ≈40% in a realistic silicone model of the middle and anterior cerebral arteries. By targeting NET-rich thrombi with piperaquine-coated retrieval devices, this strategy converts thromboinflammatory resistance into a mechanical advantage, potentially increasing first-pass success and minimizing distal embolization. Moreover, a broader panel of structurally diverse, clinically approved DNA-binding drugs, such as quinacrine, chloroquine, and amodiaquine, exhibiting similar NET-binding potential, exerts similar clot-adhesive properties in preclinical assessments (patent WO2021239905A1). This approach enables safe drug repurposing and offers a rapid translational route for NET-targeted device innovation. These results support the clinical translation of thromboinflammation-informed device design and underscore the importance of targeting thrombus biology—not only for pharmacological therapies but also for next-generation thrombectomy technologies.

Harnessing Biomimetic Device Coatings to Curb Nonembolic Thromboinflammation

Surface engineering of reconstructive devices provides another avenue to mitigate procedure-induced thromboinflammation. Beyond clot retrieval, endovascular devices used for vascular reconstruction—such as intracranial stents and flow diverters—also trigger thromboinflammatory responses due to their direct contact with circulating blood and penetration of the endothelial layer (ie, microdissections).

To minimize this iatrogenic response, self-expanding nitinol stents like the SubMax Stent (Ceroflo) are being developed to reduce mechanical trauma compared with balloon-mounted stents, especially in fragile intracranial arteries. Moreover, biomimetic surface modifications offer a promising strategy to control thromboinflammation at the device–tissue interface.

Notably, 2 complementary approaches involving CD31-mimetic peptides have shown efficacy in preclinical models. One strategy sustains CD31 signaling, actively promoting endothelial quiescence and reducing leukocyte/platelet adhesion, thereby limiting thrombus formation and chronic neointima development.[81] A second approach camouflages the device surface by mimicking the molecular landscape of healthy endothelium. This stealth strategy renders the implant immunologically silent, delivering a leave-me-alone signal to circulating platelets and leukocytes and hence preventing the foreign body response.[82] Together, these CD31-based coatings exemplify precision-engineered implants that integrate into the vascular environment while evading innate immune surveillance.

Anti-Inflammatory Approaches to Prevent Ischemia-Reperfusion Injury

Because massive free radical production during reperfusion is a key driver of neuroinflammation, intra-arterial administration of potent free radical scavengers, such as edavarone and uric acid, could significantly reduce ischemia-reperfusion injury (Table). Edaravone has demonstrated both safety and clinical benefits when administered within 48 hours of stroke symptom onset.[83] Yet, its effect in the hyperacute phase (ie, immediately after endovascular thrombectomy-mediated recanalization) remains unexplored. A randomized trial (URICO-ICTUS [Efficacy Study of Combined Treatment With Uric Acid and rtPA in Acute Ischemic Stroke; URL: https://clinicaltrials.gov, Unique identifier: NCT00860366]) evaluated the intravenous administration of uric acid in combination with alteplase and showed a nonsignificant trend toward improved functional outcomes.[84] Importantly, subgroup analysis revealed significant benefit in 2 populations: patients with hyperglycemia (where oxidative stress and neuroinflammation are amplified)[84] and patients who underwent mechanical thrombectomy, suggesting that uric acid may be particularly effective in mitigating reperfusion injury in contexts of high oxidative burden.[85] NET-targeting strategies (discussed above) may also prevent ischemia-reperfusion injury.[86] Acute adjunctive anti-inflammatory interventions may improve recanalization rates and prevent secondary complications, including early reocclusion and ischemia-reperfusion injury.

Several other studies have investigated immunomodulatory treatments in the early phase of acute ischemic stroke (within 12 hours of symptom onset), mainly targeting secondary inflammatory-mediated injury.[87] Notably, some of these inflammatory targets overlap with thromboinflammation pathways.

Blood and Imaging Biomarker-Guided Precision Medicine

The multicomponent inflammatory architecture of ischemic stroke—encompassing NETs, complement activation, VWF networks, and vessel wall inflammation—demands precision medicine strategies in which biomarker-guided approaches help optimize therapy, predict resistance, and identify patients suited for targeted inflammatory interventions.

Blood-based and imaging biomarkers now provide real-time assessment of key inflammatory mechanisms

Table. Summary of Thromboinflammatory Challenges and Therapeutic Opportunities in Acute Ischemic Stroke

Pathophysiological challenges	Molecular targets	Therapeutic opportunity	Mechanism of action	Clinical studies
Recanalization resistance and early reocclusion*	NETs	DNase-1 (dornase alfa)	Enzymatic degradation of NET DNA scaffolds, disrupting thrombus stability and enhancing fibrinolytic penetration	Phase 2 RCTs of dornase alfa: ongoing (EXTEND-IA DNase: URL: https://www.clinicaltrials.gov; Unique identifier: NCT05203224; NETs-target: URL: https://www.clinicaltrials.gov; Unique identifier: NCT04785066)
	NLRP3 inflammasome	Colchicine	Inhibition of inflammasome assembly and IL-1β/IL-18 maturation, reducing inflammatory cytokine release	Not explored in the acute phase
	Cytokines and complement	Monoclonal anticytokine antibodies (anakinra)	Blockade of proinflammatory cytokine receptors (IL-1R) or complement cascade activation	Phase 2 RCT of IL-1R antagonist (SCIL-STROKE) showed safety and tolerability
	NETs	DNA-binding compound-coated retrieval devices (piperaquine-modified stent retrievers)	DNA-binding compounds on device surfaces increase clot engagement via NET scaffold adhesion	Evidence of increased retrieval in experimental studies (Skarbek et al[80] JNIS 2023)
No-reflow phenomenon and reperfusion injury	NETs	DNase-1 (dornase alfa)	Enzymatic degradation of NETs in microvascular compartment, reducing capillary obstruction and inflammatory cascades	Phase 2 RCTs of dornase alfa: ongoing (EXTEND-IA DNase: URL: https://www.clinicaltrials.gov; Unique identifier: NCT05203224; NETs-target: URL: https://www.clinicaltrials.gov; Unique identifier: NCT04785066)
	NLRP3 inflammasome	Colchicine	Suppression of microvascular inflammation and endothelial activation	Not explored in the acute phase
	ROS	Edaravone	Free radical scavenging prevents oxidative mitochondrial damage and endothelial dysfunction	Phase 3 completed RCT of edaravone vs placebo administered within 48 h from symptom onset showed better functional outcomes (TASTE: URL: https://www.clinicaltrials.gov; Unique identifier: NCT02430350; TASTE-SL: NCT04950920; TASTE-2: URL: https://www.clinicaltrials.gov; NCT05249920)
	ROS	Uric acid	Free radical scavenging prevents oxidative mitochondrial damage and endothelial dysfunction	Phase 2b/3 RCT comparing uric acid vs placebo in adjunction to systemic thrombolysis showed no overall benefit but subgroup benefits in hyperglycemic patients and those undergoing mechanical thrombectomy (URICO-ICTUS: URL: https://www.clinicaltrials.gov; Unique identifier: NCT00860366)

EXTEND-IA DNase indicates Improving Early Reperfusion With Adjuvant Dornase Alfa in Large Vessel Ischemic Stroke; IL, interleukin; IL-1R, interleukin 1 receptor; NET, neutrophil extracellular trap; NETs-target, Efficacy of Pulmozyme on Arterial Recanalization in Post-Thrombectomy Patients Managed for Ischemic Stroke; NLRP3, NOD-, LRR- and pyrin domain-containing protein 3; RCT, randomized controlled trial; ROS, reactive oxygen species; TASTE, Study of Compound Edaravone Injection for Treatment of Acute Ischemic Stroke; TASTE2, Treatment of Acute Ischemic Stroke With Edaravone Dexborneol II; TASTE-SL, Phase III Clinical Trial of Y-2 Sublingual Tablets in the Treatment of Acute Ischemic Stroke; and URICO-ICTUS, Efficacy Study of Combined Treatment With Uric Acid and rtPA in Acute Ischemic Stroke.

*Pharmacological thrombolysis and mechanical thrombectomy.

underlying the recanalization-reperfusion gap, such as microvascular plugging, NET burden, complement activation, and reperfusion injury. This biomarker-guided approach supports personalized treatment strategies tailored to individual inflammatory profiles rather than uniform protocols.

Validated biomarkers include the following:

- MMP-9 (matrix metalloproteinase-9): Elevated levels predict blood-brain barrier disruption, hemorrhagic transformation risk, and microvascular dysfunction after recanalization[88]
- Thioredoxin: An antioxidant enzyme reflecting mitochondrial dysfunction and oxidative stress that drives reperfusion injury and infarct expansion.[89]
- Circulating cell-free DNA and myeloperoxidase-DNA complexes: Direct NET burden quantification that identifies patients likely to benefit from DNase therapy and predicts thrombectomy resistance.[90]
- Complement activation products (C3a, C5a): Indicators of microvascular inflammation that identify candidates for complement inhibition and predict postrecanalization outcomes.[58]

Advanced machine learning integration combining age, National Institutes of Health Stroke Scale score, and multiplex inflammatory panels achieves >90% accuracy in predicting futile recanalization and enables real-time decision support for therapy selection and timing of inflammatory intervention.[91] This represents a paradigm shift from empirical treatment to biomarker-informed precision stroke medicine.

Recent advances in neuroimaging enable real-time assessment of both inflammatory activity and microvascular reperfusion, offering new opportunities to guide therapy beyond conventional angiographic end points. Flat-panel computed tomography perfusion and arterial spin labeling magnetic resonance imaging can detect capillary no-reflow—missed by angiography—and outperform conventional Thrombolysis in Cerebral Infarction scores in predicting tissue-level outcomes.[92] Integration of advanced perfusion imaging into endovascular workflows could soon support real-time decisions such as intra-arterial anti-inflammatory agent delivery or additional aspiration maneuvers in persistent no-reflow scenarios.

CONCLUSIONS

The emerging science of thromboinflammation fundamentally challenges the angiographic paradigm that has dominated stroke care for decades. While achieving successful macrovascular recanalization (ie, Thrombolysis in Cerebral Infarction score ≥2b) in ≈80% to 90% of patients represents a remarkable technical achievement, the persistent gap between angiographic vessel patency and functional recovery underscores critical biological mechanisms that demand targeted therapeutic solutions.

Thromboinflammation components, such as NETs, complement activation, and vessel wall inflammation, are measurable, targetable biological processes directly contributing to ischemic injury and may help close this gap. Biomarker-guided therapy selection is moving from research to clinical reality, with machine learning algorithms achieving >90% accuracy in predicting futile recanalization. DNase trials show safety and biological activity, complement inhibitors show promise for microvascular protection, and surface-modified devices are entering clinical testing. Together, these provide proof-of-concept that biological insights can drive therapeutic innovation (Table).

For practicing clinicians, targeting thromboinflammation will likely be the focus of next-generation acute stroke therapeutics. Although these strategies are not yet part of routine clinical practice, ongoing trials testing NET-degrading enzymes, complement inhibitors, and biomimetic device coatings signal an imminent paradigm shift. Trials designed around thromboinflammatory endotypes and targets are now warranted to test mechanical and pharmacological immunomodulatory strategies in ischemic stroke.

ARTICLE INFORMATION

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